Project Initialization and Planning Phase

Date	10 July 2026
Team ID	Xxxxxx
Project Title	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The objective of this project is to develop an AI-based Flood Prediction System that predicts flood risk using historical and environmental data, helping authorities and citizens take preventive actions.
Scope	The system collects flood-related data, preprocesses it, trains a machine learning model, predicts flood occurrence, and displays results through a simple dashboard for decision-making.
Problem Statement	
Description	Floods cause severe damage to life and property. Existing warning systems may not provide timely or accurate predictions. This project improves flood prediction using machine learning techniques.
Impact	The proposed solution helps reduce flood-related losses, supports disaster management, improves early warning systems, and assists government authorities and the public in taking preventive measures.
Proposed Solution	
Approach	The project uses data preprocessing, feature selection, machine learning algorithms, model evaluation, and prediction to identify flood-prone areas accurately.
Key Features	The project uses data preprocessing, feature selection, machine learning algorithms,

	model evaluation, and prediction to identify flood-prone areas accurately.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5 / Google Colab CPU (Optional T4 GPU)
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	10 GB free Storage / SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, Google Colab, VS Code
Data		
Data	Source, size, format	Flood Prediction Dataset(CSV),Kaggle Dataset